



US012413114B2

(12) **United States Patent**  
**Kobayashi**

(10) **Patent No.:** **US 12,413,114 B2**  
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **VIBRATION MOTOR WITH HOUSING  
HAVING FIRST AND SECOND NOTCHES  
FOR LEAD WIRE AND FLEXIBLE PRINTED  
CIRCUIT BOARD**

(71) Applicant: **Nidec Corporation**, Kyoto (JP)

(72) Inventor: **Tomohiro Kobayashi**, Kyoto (JP)

(73) Assignee: **NIDEC CORPORATION**, Kyoto (JP)

(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 244 days.

(21) Appl. No.: **18/127,799**

(22) Filed: **Mar. 29, 2023**

(65) **Prior Publication Data**

US 2023/0318396 A1 Oct. 5, 2023

(30) **Foreign Application Priority Data**

Mar. 31, 2022 (JP) ..... 2022-058224

(51) **Int. Cl.**

**H02K 33/02** (2006.01)

**H02K 5/22** (2006.01)

(52) **U.S. Cl.**

CPC ..... **H02K 5/225** (2013.01); **H02K 33/02**  
(2013.01); **H02K 2203/03** (2013.01)

(58) **Field of Classification Search**

CPC .. H02K 7/1869; H02K 7/1876; H02K 7/1892;  
H02K 33/12; H02K 33/00; H02K 33/02;  
H02K 33/04; H02K 33/06; H02K 33/16;  
H02K 35/00; H02K 35/02; H02K 5/02;  
H02K 5/04; H02K 5/24; H02K 5/22;  
H02K 5/225; H02K 2203/03

USPC ..... 310/15–25, 89, 71

See application file for complete search history.

(56) **References Cited**

**U.S. PATENT DOCUMENTS**

1,918,364 A \* 7/1933 Winsor ..... H02K 33/12  
310/35  
4,529,906 A \* 7/1985 McMahon ..... G01S 1/72  
310/27  
5,107,155 A \* 4/1992 Yamaguchi ..... H02K 7/063  
340/407.1  
5,661,352 A \* 8/1997 Oguchi ..... H02K 5/225  
310/71  
5,705,866 A \* 1/1998 Oguchi ..... H02K 1/187  
310/90  
5,730,619 A \* 3/1998 Hamlin ..... H01R 13/193  
439/573  
6,121,701 A \* 9/2000 Kloeppel ..... G11B 19/2009  
310/71  
6,271,610 B1 \* 8/2001 Ibata ..... H02K 5/00  
310/40 MM  
6,413,117 B1 \* 7/2002 Annerino ..... H01R 12/7076  
439/500

(Continued)

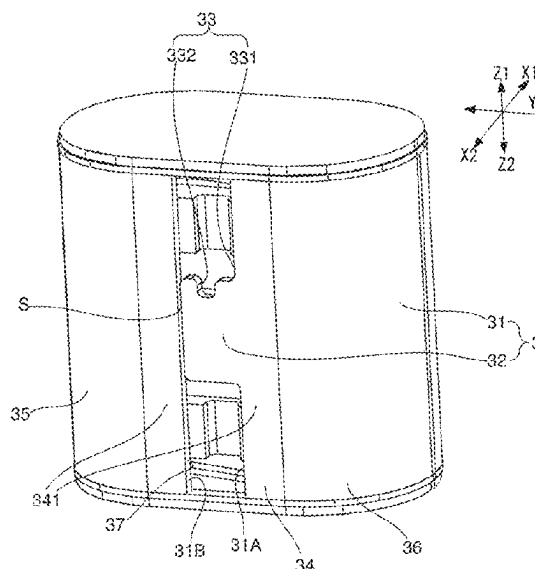
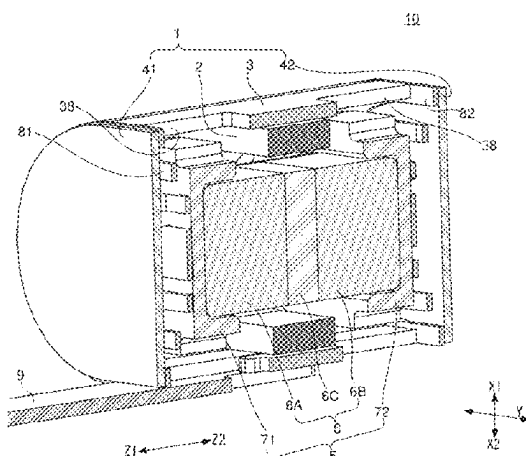
*Primary Examiner* — Maged M Almawri

(74) *Attorney, Agent, or Firm* — Keating & Bennett, LLP

(57) **ABSTRACT**

A vibration motor includes a stationary portion and a movable portion to vibrate in a center axis direction with respect to the stationary portion. The stationary portion includes a coil to apply a driving force to the movable portion by energization, and a housing that accommodates the movable portion and the coil. The housing includes a notch recessed inward from an outer edge of the housing in the center axis direction. The notch includes a first notch recessed inward from an outer edge of the housing in the center axis direction, and a second notch recessed inward from an outer edge of the first notch in the center axis direction. A lead wire of the coil extends out of the housing through the notch.

**8 Claims, 7 Drawing Sheets**



# US 12,413,114 B2

Page 2

|      |                         |                        |                     |         |                   |                        |
|------|-------------------------|------------------------|---------------------|---------|-------------------|------------------------|
| (56) | <b>References Cited</b> |                        | 2008/0291650 A1 *   | 11/2008 | Hautvast .....    | H05K 1/147<br>361/772  |
|      | U.S. PATENT DOCUMENTS   |                        | 2009/0039721 A1 *   | 2/2009  | Takagi .....      | H05K 3/341<br>310/81   |
|      | 6,737,780 B1 *          | 5/2004 Fisher .....    | 2010/0102645 A1 *   | 4/2010  | Trietz .....      | H02N 2/04<br>310/12.31 |
|      | 7,579,731 B2 *          | 8/2009 Fukushima ..... | 2011/0291510 A1 *   | 12/2011 | Hara .....        | H02K 7/061<br>310/91   |
|      | 7,646,122 B2 *          | 1/2010 Uchiumi .....   | 2013/0169071 A1 *   | 7/2013  | Endo .....        | H02K 33/12<br>310/25   |
|      | 7,825,555 B2 *          | 11/2010 Uchiumi .....  | 2014/0035397 A1 *   | 2/2014  | Endo .....        | H02K 33/18<br>310/30   |
|      | 8,004,131 B2 *          | 8/2011 Yu .....        | 2014/0062225 A1 *   | 3/2014  | Kim .....         | H02K 33/00<br>310/15   |
|      | 9,742,240 B2 *          | 8/2017 Katada .....    | 2014/0084710 A1 *   | 3/2014  | Endo .....        | H02K 33/16<br>310/25   |
|      | 10,111,324 B2 *         | 10/2018 Koepsell ..... | 2014/0175955 A1 *   | 6/2014  | Katada .....      | H02K 5/225<br>310/68 R |
|      | 10,890,169 B2 *         | 1/2021 Jeong .....     | 2014/0306556 A1 *   | 10/2014 | Kim .....         | H02K 33/16<br>310/25   |
|      | 11,418,099 B2 *         | 8/2022 Takahashi ..... | 2015/0123498 A1 *   | 5/2015  | Yang .....        | H02K 33/16<br>310/25   |
|      | 2001/0013730 A1 *       | 8/2001 Yamaguchi ..... | 2015/0372569 A1 *   | 12/2015 | Lau .....         | H02K 5/148<br>310/71   |
|      | 2002/0140304 A1 *       | 10/2002 Dreher .....   | 2018/0219465 A1 *   | 8/2018  | Katada .....      | B06B 1/045             |
|      | 2003/0127921 A1 *       | 7/2003 Akutsu .....    | 2019/0070635 A1 *   | 3/2019  | Takeda .....      | H02K 33/18             |
|      | 2004/0256920 A1 *       | 12/2004 Gummin .....   | 2019/0115816 A1 *   | 4/2019  | Takahashi .....   | A45D 29/14             |
|      | 2005/0046302 A1 *       | 3/2005 Suzuki .....    | 2019/0305659 A1 *   | 10/2019 | Kitahara .....    | H02K 33/18             |
|      | 2005/0073205 A1 *       | 4/2005 Takagi .....    | 2019/0379263 A1 *   | 12/2019 | Zhu .....         | H02K 33/06             |
|      | 2006/0138885 A1 *       | 6/2006 Uchiumi .....   | 2020/0076287 A1 *   | 3/2020  | Takahashi .....   | H02K 33/06             |
|      | 2006/0170296 A1 *       | 8/2006 Nakajima .....  | 2020/0161955 A1 *   | 5/2020  | Kitahara .....    | H02K 33/16             |
|      |                         |                        | 2021/0336521 A1 *   | 10/2021 | Ando .....        | H02K 33/18             |
|      |                         |                        | 2023/0198366 A1 *   | 6/2023  | Tsuchihashi ..... | H02K 33/18<br>310/28   |
|      |                         |                        | * cited by examiner |         |                   |                        |

Fig. 1

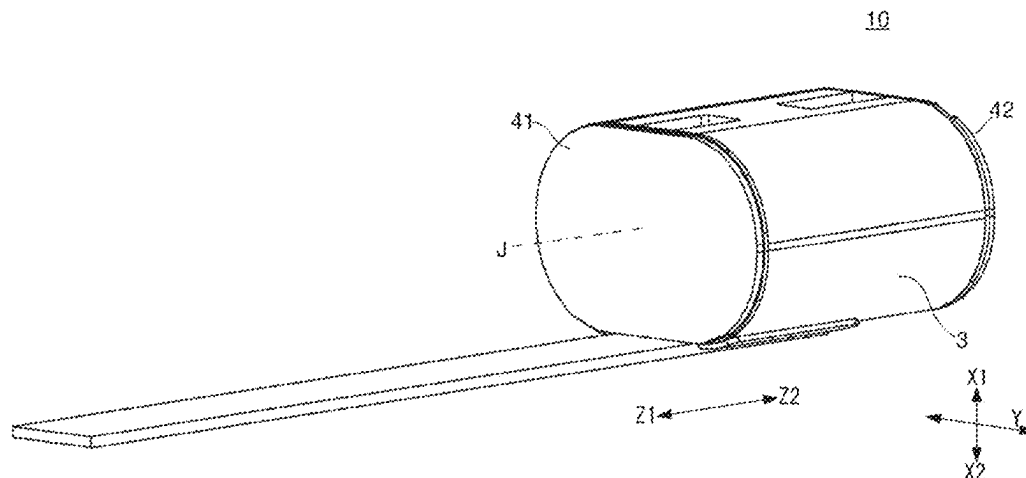


Fig. 2

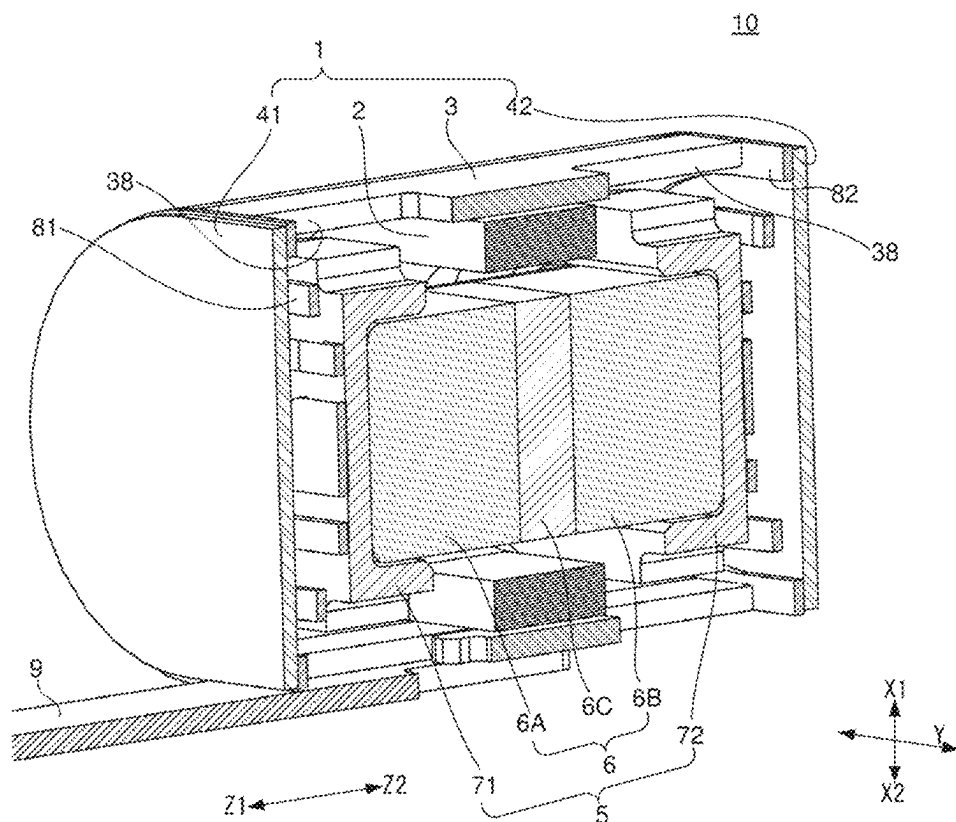


Fig. 3

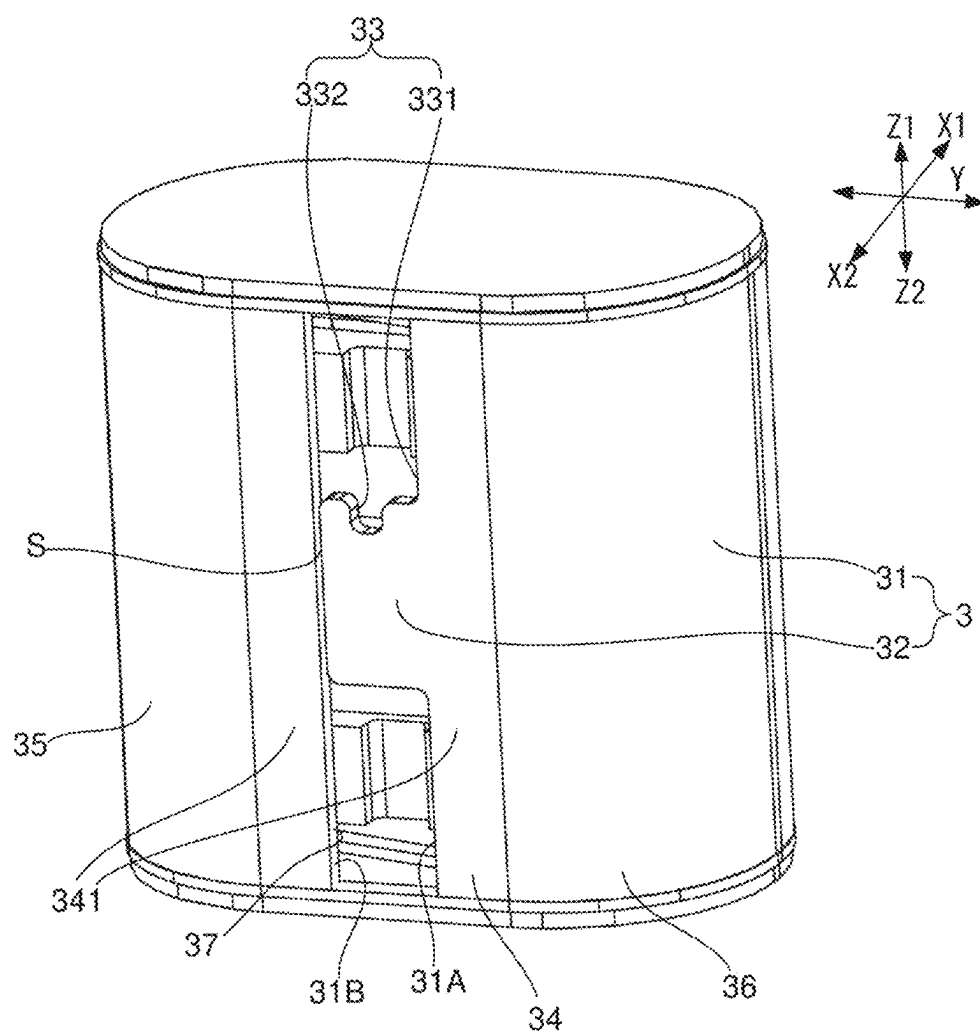


Fig. 4

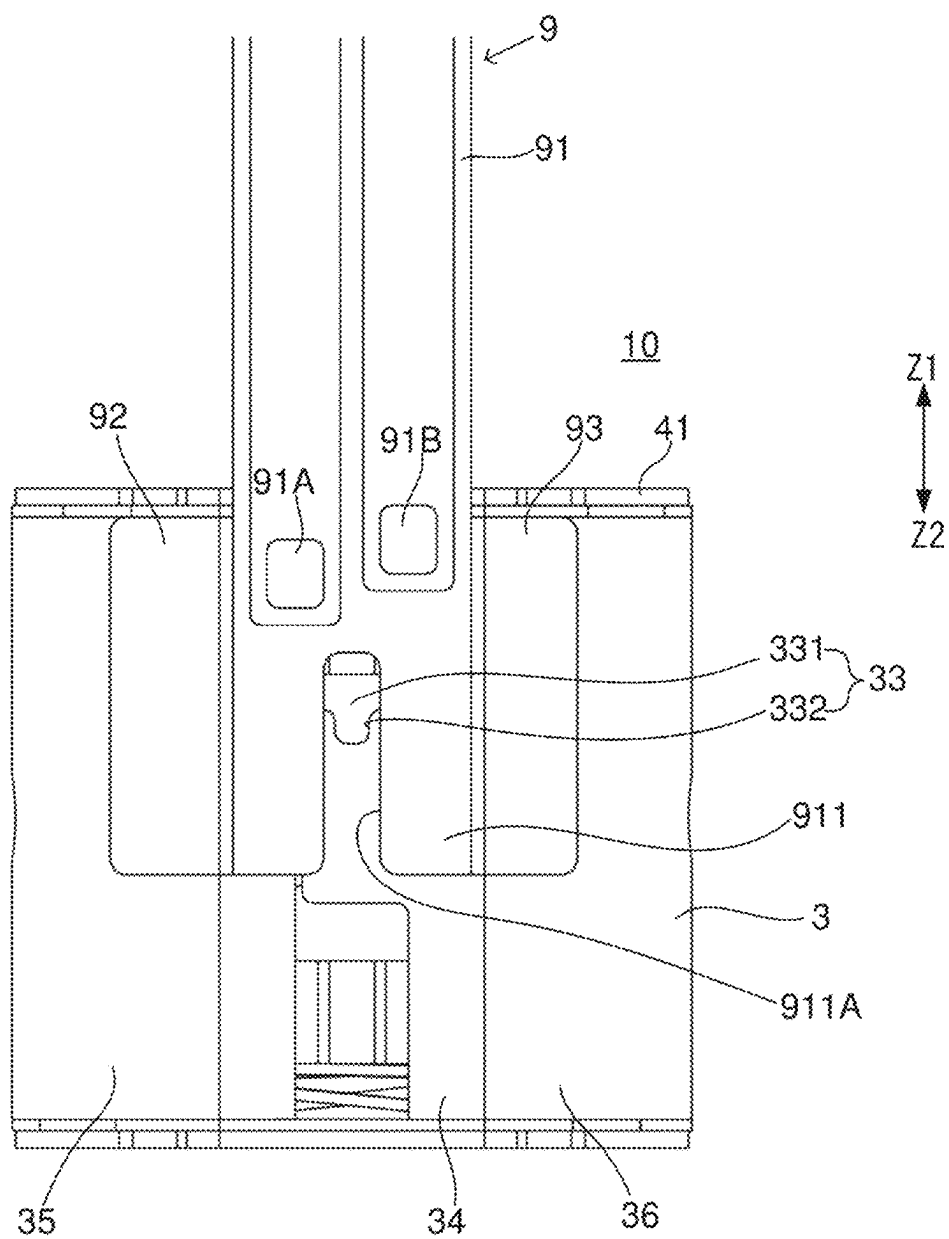


Fig. 5

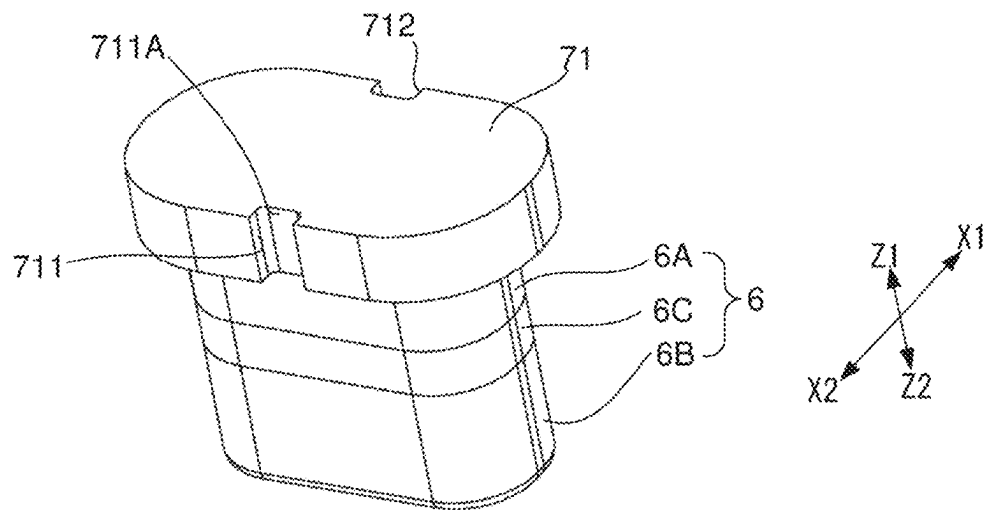


Fig. 6

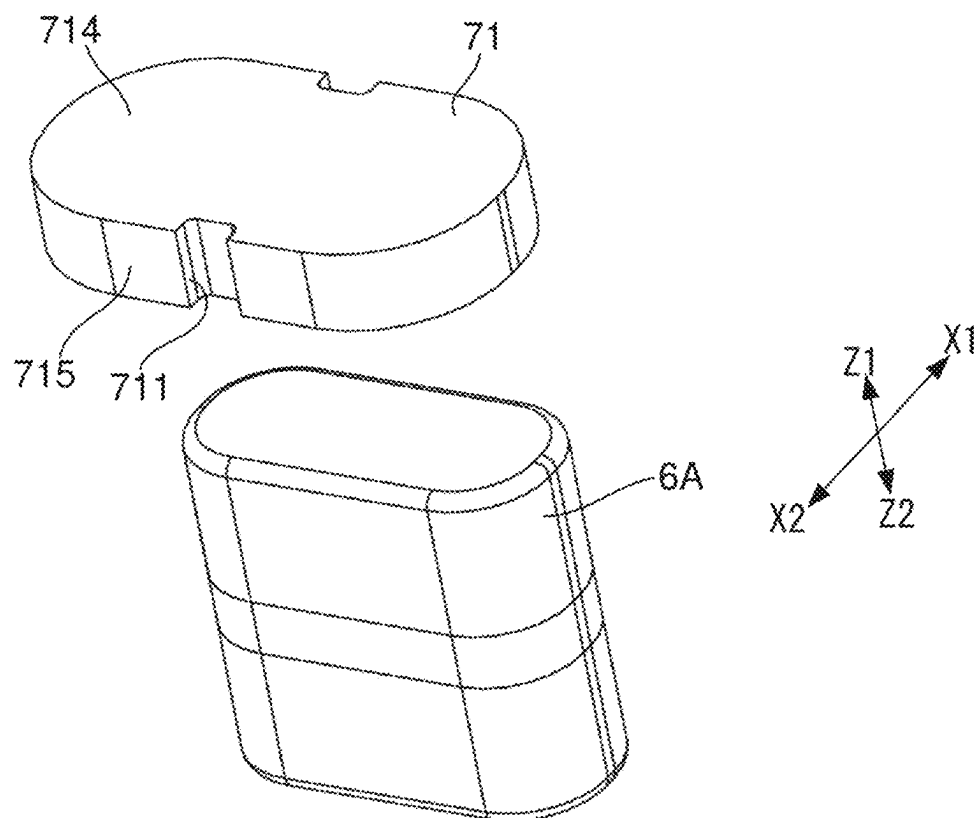


Fig. 7

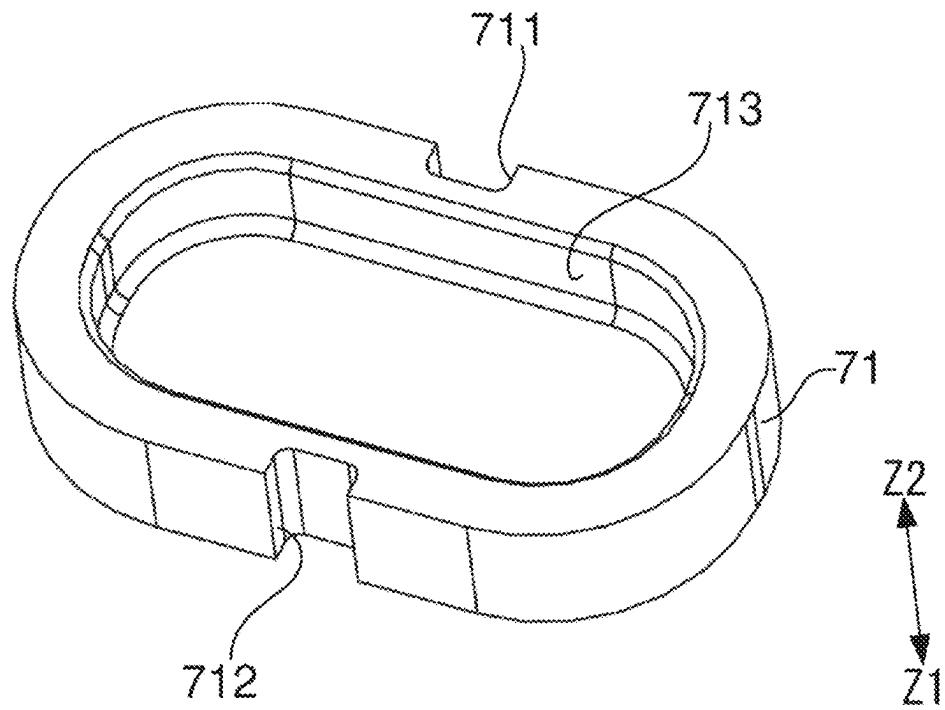


Fig. 8

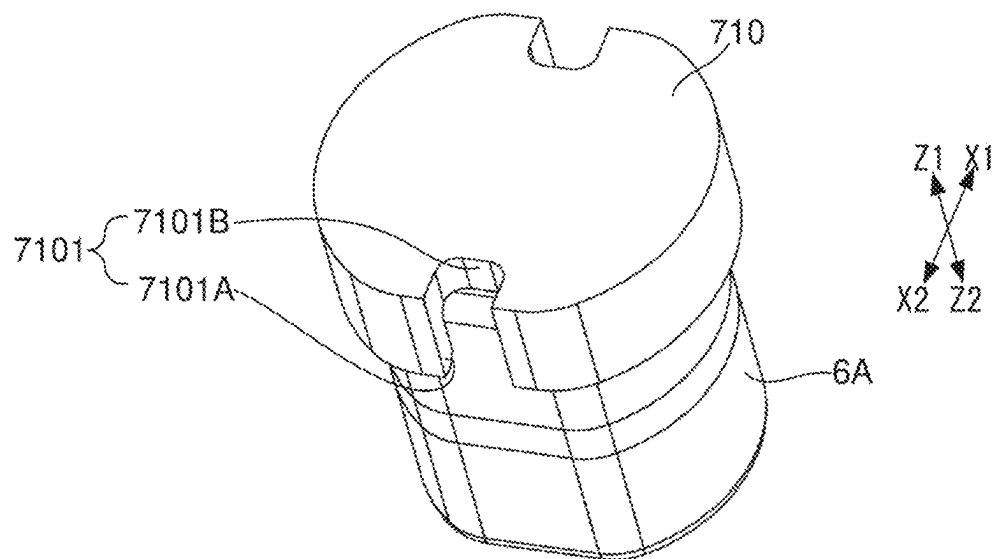


Fig. 9

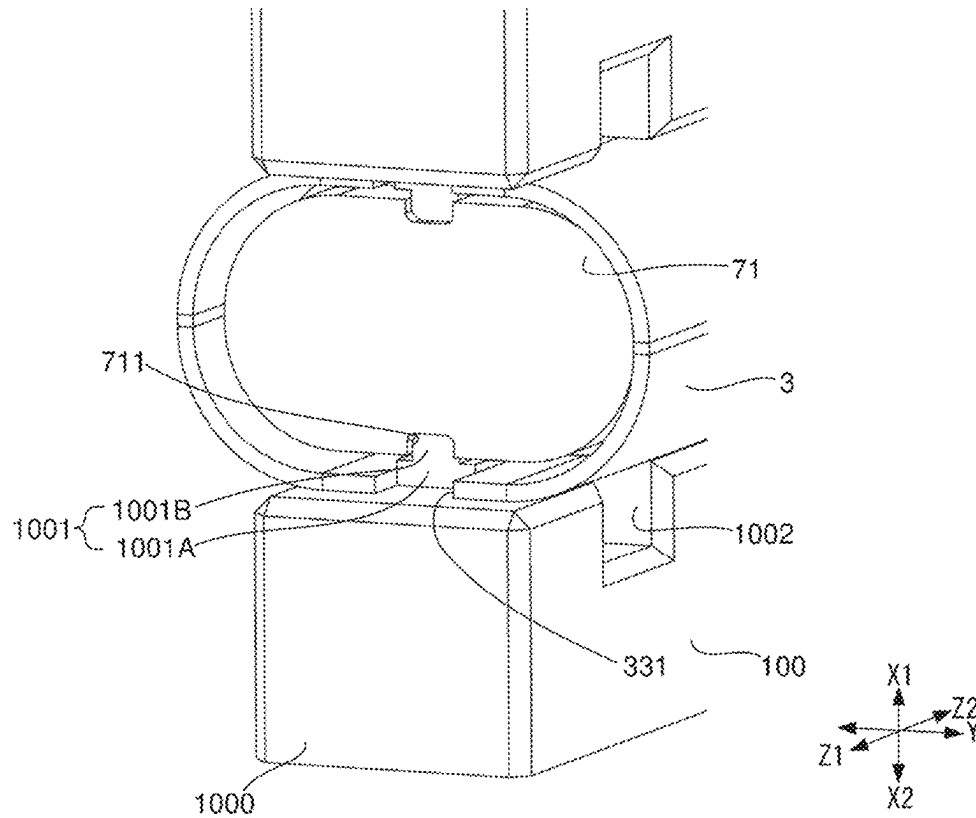


Fig. 10

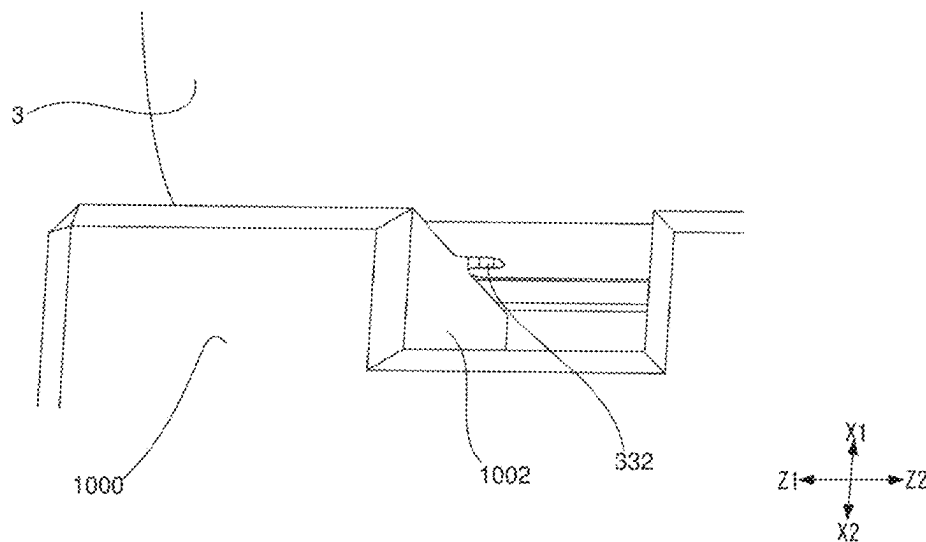
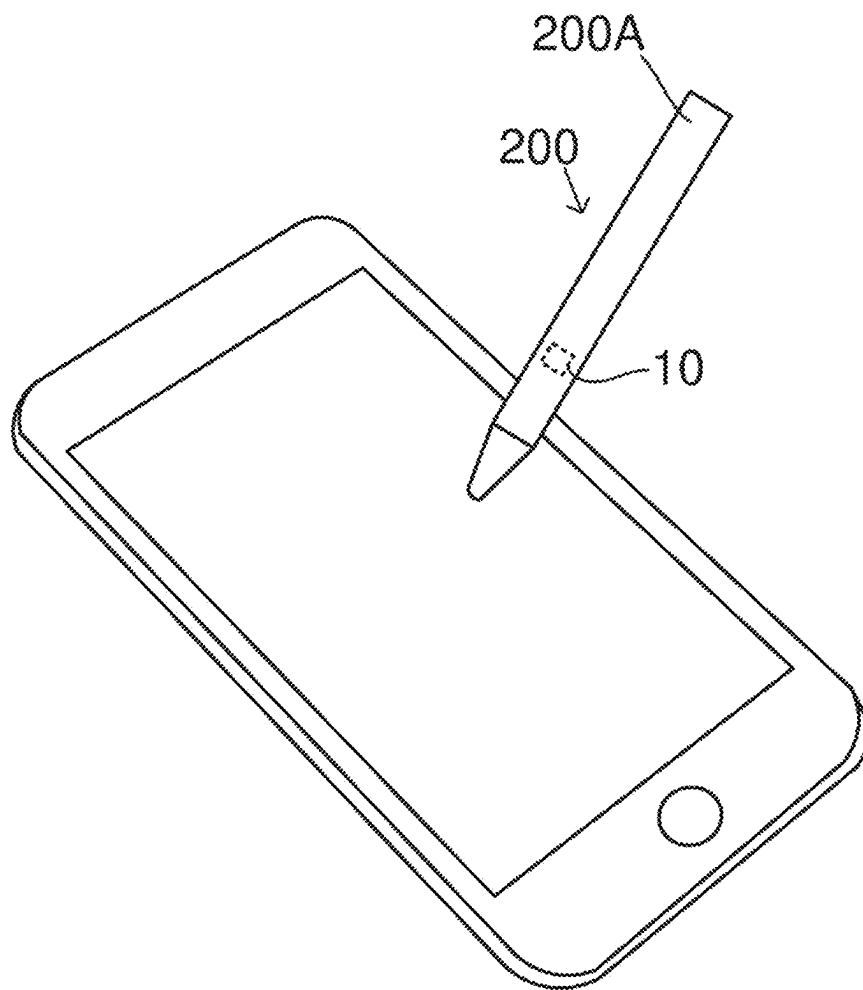




Fig. 11



1

# VIBRATION MOTOR WITH HOUSING HAVING FIRST AND SECOND NOTCHES FOR LEAD WIRE AND FLEXIBLE PRINTED CIRCUIT BOARD

## CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2022-058224, filed on Mar. 31, 2022, the entire contents of which are hereby incorporated herein by reference.

## 1. FIELD OF THE INVENTION

The present disclosure relates to a vibration motor.

## 2. BACKGROUND

Conventionally, various apparatuses such as a smartphone and other portable devices include a vibration motor as a vibration generation device. The vibration motor is used for a function of notifying the user of an incoming call, an alarm, and the like, or a function of haptic feedback in a human interface, for example.

In general, a vibration motor includes a stator, an elastic member, and a vibrator. The stator includes a chassis and a coil. The vibrator includes a magnet. The vibrator and the chassis are connected by an elastic member. When the coil is energized to generate a magnetic field, the vibrator vibrates. However, in the conventional vibration motor, a circuit board and a coil are connected inside a chassis, and there is a problem in connection workability.

## SUMMARY

An example embodiment of a vibration motor of the present disclosure includes a stationary portion and a movable portion to vibrate in a center axis direction with respect to the stationary portion. The stationary portion includes a coil to apply a driving force to the movable portion by energization, and a housing that accommodates the movable portion and the coil. The housing includes a notch recessed inward from an outer edge of the housing in the center axis direction. The notch includes a first notch recessed inward from an outer edge of the housing in the center axis direction, and a second notch recessed inward from an outer edge of the first notch in the center axis direction. A lead wire of the coil extends out of the housing through the notch.

The above and other elements, features, steps, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view illustrating an external appearance of a vibration motor according to an example embodiment of the present disclosure.

FIG. 2 is a cross-sectional perspective view of the vibration motor.

FIG. 3 is a perspective view of a partial configuration of the vibration motor as viewed from a first region side.

FIG. 4 is a plan view illustrating a partial configuration of the vibration motor.

2

FIG. 5 is a perspective view illustrating a configuration of a core portion and a holder according to an example embodiment of the present disclosure in a movable portion.

FIG. 6 is an exploded perspective view in which a core portion and a holder according to an example embodiment of the present disclosure are disassembled.

FIG. 7 is a perspective view of a holder according to an example embodiment of the present disclosure.

FIG. 8 is a perspective view illustrating a configuration of a holder according to a comparative example embodiment of the present disclosure.

FIG. 9 is a perspective view illustrating the middle of the manufacturing process of the vibration motor according to an example embodiment of the present disclosure.

FIG. 10 is a perspective view of a housing according to an example embodiment of the present disclosure as viewed through a hole of a jig.

FIG. 11 is a diagram illustrating an example of a haptic device according to an example embodiment of the present disclosure.

## DETAILED DESCRIPTION

Hereinafter, example embodiments of the present disclosure will be described with reference to the drawings.

In the drawings, a center axis direction in which the center axis J of a vibration motor 10 extends is defined as a Z direction, one side in the center axis direction is defined as Z1, and the other side in the center axis direction is defined as Z2. Additionally, a first direction perpendicular to the center axis direction is defined as an X direction, and one side in the first direction is defined as X1, while the other side in the first direction is defined as X2. Additionally, a second direction perpendicular to the center axis direction and the first direction is referred to as a Y direction.

FIG. 1 is a perspective view illustrating an external appearance of the vibration motor 10 according to the present example embodiment of the present disclosure.

FIG. 2 is a cross-sectional perspective view of the vibration motor 10.

The vibration motor 10 includes a stationary portion 1, a movable portion 5, and elastic members 81 and 82. The movable portion 5 is capable of vibrating or in the center axis direction (Z direction) with respect to the stationary portion 1.

The stationary portion 1 includes a coil 2, a housing 3, and lids 41 and 42. The housing 3 is a plate-like member formed to extend in the circumferential direction around the center axis J as the circumferential direction. The housing 3 has a tubular shape extending in the center axis direction. The housing 3 is made of a magnetic body.

The magnetic body is stainless steel, for example.

The coil 2 is formed by winding a conductive wire around the center axis J, and is fixed to the inner surface of the housing 3. That is, the housing 3 accommodates the coil 2 therein. The coil 2 generates a magnetic field by being energized. The coil 2 is disposed at a center portion in the center axis direction of the housing 3.

The movable portion 5 includes a core portion 6 and holders 71 and 72, and is accommodated inside the housing 3. That is, the stationary portion 1 includes the housing 3 that accommodates the movable portion 5 and the coil 2.

The core portion 6 includes a magnet 6A on one side in the center axis direction and a magnet 6B on the other side in the center axis direction. The core portion 6 further includes a magnetic body portion 6C. The magnetic body

3

portion 6C is disposed to be held between the magnets 6A and 6B in the center axis direction.

For example, the other side in the center axis direction of the magnet 6A is the N pole, and one side in the center axis direction is the S pole. One side in the center axis direction of the magnet 6B is the N pole, and the other side in the center axis direction is the S pole. Thus, the N poles oppose each other in the center axis direction across the magnetic body portion 6C. That is, the magnetic poles of the magnets 6A and 6B on the side facing the magnetic body portion 6C are the same. As a result, the magnetic flux flows from the S pole to the N pole in the magnets 6A and 6B, flows radially outward in the magnetic body portion 6C, and penetrates the coil 2 in the radial direction. Note that the radial direction is a radial direction with respect to the center axis J. With the housing 3 made of a magnetic body, the magnetic flux having penetrated the coil 2 flows in the housing 3 in the center axis direction and is returned to the S poles of the magnets 6A and 6B. With such a configuration, the amount of magnetic flux penetrating the coil 2 can be increased.

The N pole and the S pole of the magnet may be reversed. In this case, the magnetic flux flows in a direction opposite to the above.

The holder 71 holds one end portion in the center axis direction of the magnet 6A. The holder 72 holds the other end portion in the center axis direction of the magnet 6B. The holders 71 and 72 function as weights, and are made of a tungsten alloy, for example.

The elastic members 81 and 82 are compression springs that can expand and contract in the center axis direction. The other end portion in the center axis direction of the elastic member 81 is fixed to the end face on one side in the center axis direction of the holder 71. The one end portion in the center axis direction of the elastic member 81 is fixed to the end face on one side in the center axis direction of the housing 3.

The one end portion in the center axis direction of the elastic member 82 is fixed to the end face on one side in the center axis direction of the holder 72. The other end portion in the center axis direction of the elastic member 82 is fixed to the end face on the other side in the center axis direction of the housing 3.

The lids 41 and 42 are plate-like members whose thickness direction is the center axis direction. The lid 41 is disposed on one side in the center axis direction of the elastic member 81, and is fixed to one end portion in the center axis direction of the elastic member 81. The lid 42 is disposed on the other side in the center axis direction of the elastic member 82, and is fixed to the other end portion in the center axis direction of the elastic member 82.

In a non-operational state in which the coil 2 is not energized, the movable portion 5 is in a stopped state by application of the elastic force toward the other side in the center axis direction by the elastic member 81 compressed from the natural length state and the elastic force toward the one side in the center axis direction by the elastic member 82 compressed from the natural length state. In the non-operational state, the magnetic body portion 6C of the movable portion 5 is located at the center position of the coil 2 in the center axis direction (FIG. 2). The core portion 6 is disposed radially inside the coil 2.

When the coil 2 is energized, driving force is applied to the core portion 6 by an interaction between the magnetic field generated by the coil 2 and the magnetic field generated by the core portion 6. That is, the stationary portion 1 includes the coil 2 that applies a driving force to the movable

4

portion 5 by energization. When the driving force is applied to the movable portion 5, the movable portion 5 vibrates in the center axis direction.

Thus, since the vibration motor 10 can be configured not to use a shaft that guides the movable portion unlike the conventional case, it is possible to suppress reduction in the weight of the movable portion 5 and the volume of the core portion 6. Accordingly, the amount of vibration of the vibration motor 10 can be secured. Sliding between the movable portion and the shaft is eliminated, and noise and wear due to sliding are avoided.

FIG. 3 is a perspective view of a partial configuration of the vibration motor 10 as viewed from a first region 34 side. As illustrated in FIG. 3, the housing 3 includes a continuous portion 31 formed continuously in the circumferential direction, and a protrusion 32 protruding to one side in the circumferential direction from a one circumferential end portion 31A of the continuous portion 31. The protrusion 32 has a gap S with another circumferential end portion 31B of the continuous portion 31.

The protrusion 32 is located at the center of the housing 3 in the center axis direction. A first notch 331 is formed by being surrounded by the other circumferential end portion 31B, the one circumferential end portion 31A, and the protrusion 32. A second notch 332 is formed by being recessed from one end in the center axis direction to the other side in the center axis direction of the protrusion 32. That is, the first notch 331 is formed by the other circumferential end portion 31B, the one circumferential end portion 31A, and the protrusion 32, and the second notch 332 is provided in the protrusion 32.

Thus, the housing 3 can be manufactured by bending one plate material, and the first notch 331 and the second notch 332 can be easily formed.

The first notch 331 and the second notch 332 constitute a notch 33. The notch 33 is recessed from the one side end in the center axis direction of the housing 3 to the other side in the center axis direction. That is, the housing 3 has the notch 33 recessed inward from the outer edge of the housing 3 in the center axis direction. The notch 33 includes a first notch 331 recessed inward from the outer edge of the housing 3 in the center axis direction and a second notch 332 recessed inward from the outer edge of the first notch 331 in the center axis direction. In other words, the housing 3 includes the notch 33 recessed from the outer edge of the housing 3 toward the other side in the center axis direction with the outside of the housing 3 as one side in the center axis direction.

A lead wire (not illustrated) of the coil 2 is drawn out of the housing 3 through the notch 33. Here, FIG. 4 is a plan view illustrating a partial configuration of the vibration motor 10. FIG. 4 is a view of the housing 3 as viewed in a direction perpendicular to the first region 34. The first region 34 is a flat bottom portion disposed on the other side in the first direction in the housing 3. That is, the first region 34 is a flat portion.

As illustrated in FIG. 4, a flexible printed circuit (FPC) 9 is fixed to the housing 3. That is, the stationary portion 1 includes the FPC 9 fixed to the housing 3. The FPC 9 has a base portion 91 extending in the center axis direction. The other end portion 911 in the center axis direction of the base portion 91 is fixed to the first region 34. The base portion 91 protrudes and extends from the lid 41 to the other side in the center axis direction. That is, the FPC 9 has the base portion 91 extending in the center axis direction and fixed to the first region 34.

5

The base portion **91** has electrode portions **91A** and **91B**. The lead wire drawn out from the notch **33** is connected to the electrode portions **91A** and **91B**. Thus, in the vibration motor **10** of the present example embodiment, since the lead wire of the coil **2** is connected to the FPC **9** outside the housing **3**, the workability of connection is improved.

As illustrated in FIG. 3, the first region **34** includes the protrusion **32** and both side portions **341** in the circumferential direction of the protrusion **32**. The both side portions **341** in the circumferential direction extend in the center axis direction. Therefore, the first region **34** is provided with the first notch **331** and the second notch **332**. That is, the housing **3** has the first region **34** in which the first notch **331** and the second notch **332** are formed.

As illustrated in FIG. 4, the other end portion **911** in the center axis direction in the FPC **9** is fixed so as to cover a part of the first notch **331**. The other end portion **911** in the center axis direction may be fixed so as to cover the entire first notch **331**. That is, the FPC **9** is fixed to the outer peripheral surface of the first region **34** so as to cover at least a part of the first notch **331**. As a result, it is possible to suppress the lead wire from swinging in the notch **33**.

As illustrated in FIG. 4, the other end portion **911** in the center axis direction in the FPC **9** has an FPC notch **911A** recessed on one side in the center axis direction. That is, the FPC **9** has the FPC notch **911A** recessed from the outer edge on the other side in the center axis direction toward the one side in the center axis direction in the direction at the time of fixing to the first region **34**. The lead wire of the coil **2** is drawn out through the notch **33** and the FPC notch **911A**. That is, the lead wire of the coil **2** is drawn out to the outside of the housing **3** through the notch **33** and the FPC notch **911A**.

When fixing the FPC **9** to the housing **3**, it is possible to move the FPC **9** from one side in the center axis direction to the other side and put the lead wire into the FPC notch **911A**, so that the FPC **9** can be easily fixed. Since both side portions holding the FPC notch **911A** are fixed to the housing **3**, the fixing area is increased and the fixing can be reinforced.

As illustrated in FIG. 4, the FPC **9** is provided with a protruding piece **92** protruding from the other end portion **911** in the center axis direction to one side in the circumferential direction and a protruding piece **93** protruding from the other end portion **911** in the center axis direction to the other side in the circumferential direction. The housing **3** includes a bent portion **35** connected to one side in the circumferential direction of the first region **34** and a bent portion **36** connected to the other side in the circumferential direction of the first region **34**. The bent portion **35** and the bent portion **36** are bent so as to protrude in the second direction away from each other. The protruding piece **92** is fixed to the outer peripheral surface of the bent portion **35**. The protruding piece **93** is fixed to the outer peripheral surface of the bent portion **36**. That is, the FPC **9** has the protruding pieces **92** and **93** that protrude from the base portion **91** to both sides in the circumferential direction and are fixed to the bent portions **35** and **36**.

That is, the housing **3** has the bent portions **35** and **36** connected to both end portions in the circumferential direction of the first region **34**, and the FPC **9** is fixed to the first region **34** and the bent portions **35** and **36**. As a result, the contact area of the FPC **9** with respect to the housing **3** increases, and the fixing strength can be improved.

As illustrated in FIG. 4, when viewed in a direction perpendicular to the first region **34**, the protruding pieces **92** and **93** overlap the electrode portions **91A** and **91B** in the

6

center axis direction. As a result, when tension is applied to the base portion **91**, it is possible to suppress peeling of the portions of the base portion **91** where the electrode portions **91A** and **91B** are provided from the housing **3**, and to suppress connection failure or the like between the lead wire and the electrode portions **91A** and **91B**. For the same effect, the protruding pieces **92** and **93** may be disposed on one side in the center axis direction (above the paper surface of FIG. 4) with respect to the electrode portions **91A** and **91B** when viewed in a direction perpendicular to the first region **34**.

When the housing has a plurality of notches, the second notch may be provided in at least one notch. For example, in the present example embodiment, the housing **3** has a notch in addition to the notch **33** including the first notch **331** and the second notch **332**. The other notch has the same shape as the notch **33** except for the second notch **332**, and one notch **37** (FIG. 3) is provided on the other side in the center axis direction and two notches **38** (FIG. 2) are provided at a location of the housing **3** located on the opposite side to the first region **34** across the central axis.

FIG. 5 is a perspective view illustrating a configuration of the core portion **6** and the holder **71** in the movable portion **5**. FIG. 5 illustrates a state in which the holder **71** is attached to the core portion **6**. FIG. 6 is an exploded perspective view of the core portion **6** and the holder **71**. FIG. 7 is a perspective view of the holder **71**.

The holder **71** has a recess **713** (FIG. 7) recessed toward one side in the center axis direction. The holder **71** is fixed to the magnet **6A** by inserting the magnet **6A** into the recess **713**. That is, the movable portion **5** includes the magnet **6A** and the holder **71** fixed to one side in the center axis direction of the magnet **6A** and having a larger radial width than the magnet **6A**.

The holder **71** has a holder notch **711** that is recessed from the outer edge of the holder **71** toward one side in the first direction. A bottom portion **711A** of the holder notch **711** is disposed on the other side in the first direction with respect to the outer edge of the magnet **6A**.

Here, FIG. 8 is a perspective view illustrating a configuration of a holder **710** according to a comparative example. The holder **710** has a holder notch **7101**. The holder notch **7101** has openings **7101A** and **7101B**. The opening **7101A** opens in the first direction. The opening **7101B** is connected to one side in the center axis direction of the opening **7101A**, and opens in the center axis direction. When the opening **7101A** is viewed in the first direction, a part of the other end surface in the first direction of the magnet **6A** is exposed. When the opening **7101B** is viewed in the center axis direction, a part of the end surface on one side in the center axis direction of the magnet **6A** is exposed.

On the other hand, in the configuration according to the present example embodiment (FIG. 5), the magnet **6A** is not exposed even when the holder notch **711** is viewed in the first direction and the center axis direction. Therefore, the contact area of the holder **71** with respect to the magnet **6A** is increased, and the fixing strength of the holder **71** is improved. In the case of fixing with the liquid adhesive, the possibility that the adhesive leaks from the holder notch **711** at the time of fixing is reduced.

In addition, as illustrated in FIG. 6, the holder **71** includes a holder flat portion **714** that extends perpendicularly to the center axis J, and a holder wall portion **715** that extends from the outer end portion in the first direction of the holder flat portion **714** to the other side in the center axis direction. The holder flat portion **714** covers at least a part of the outer end portion in the first direction of the magnet **6A**. The contact area between the holder **71** and the magnet **6A** is increased

7

by the holder flat portion **714** and the holder wall portion **715**, and the fixing strength of the holder **71** can be improved.

The holder **71** includes a holder notch **712** at a position facing the holder notch **711** in the first direction.

Here, a jig used when the vibration motor **10** is manufactured will be described. FIG. **9** is a perspective view illustrating the middle of the manufacturing process of the vibration motor **10**. In FIG. **9**, the housing **3** is installed on a jig **100**.

The jig **100** includes a base portion **1000**, a positioning portion **1001**, and a hole **1002**. The positioning portion **1001** is disposed on one side in the first direction of the base portion **1000**. The positioning portion **1001** includes a first extending portion **1001A** extending in the center axis direction and a second extending portion **1001B** extending in the center axis direction. The second extending portion **1001B** is disposed on one side in the first direction of the first extending portion **1001A**. The width in the second direction of the second extending portion **1001B** is narrower than the width in the second direction of the first extending portion **1001A**. That is, the positioning portion **1001** is formed in a rail shape extending in the center axis direction.

The hole **1002** is formed to be recessed from the surface on one side in the first direction of the base portion **1000** to the other side in the first direction and penetrates in the second direction.

The first extending portion **1001A** is inserted into the first notch **331** of the housing **3**, and the housing **3** is installed on the base portion **1000**. As illustrated in FIG. **10**, in such an installation state, the hole **1002** and the second notch **332** face each other in the first direction.

The housing **3** is installed in the jig **100** in a state where the lead wire is drawn out from the second notch **332** to the outside of the housing **3** in advance. Since the first notch **331** is used for fixing the jig **100** and the second notch **332** is used for drawing the lead wire, interference between the jig and the lead wire is suppressed. Accordingly, an adverse effect on the lead wire can be suppressed.

In addition, the holder **71** fixed to the magnet **6A** is installed in the positioning portion **1001**. More specifically, the holder **71** is slid from one side in the center axis direction to the other side in the center axis direction so that the second extending portion **1001B** passes through the holder notch **711** of the holder **71**. Thus, in the manufactured vibration motor **10**, the first notch **331** overlaps the holder notch **711** when viewed from the first direction. That is, the holder **71** has the holder notch **711** overlapping the notch **33** when viewed from the first direction. Accordingly, the relative positioning between the housing **3** and the holder **71** can be performed by fixing the notch **33** of the housing **3** and the holder notch **711** to the positioning portion **1001**.

The holder notch **711** penetrates from one side in the center axis direction to the other side in the center axis direction (see FIG. **5**). Thus, the magnet **6A** can be inserted into the housing **3** by sliding the holder **71** in the center axis direction while positioning on the positioning portion **1001**.

The holder notch **711** is provided in the holder flat portion **714** and the holder wall portion **715** (see FIG. **6**). Thus, the holder notch **711** can be formed long in the center axis direction, and the holder **71** can be stably installed on the jig **100**.

The vibration motor **10** according to the above-described example embodiment can be mounted on various electronic devices.

8

By the above, an electronic device can be vibrated to realize functions such as notification to the operator or tactile feedback.

The vibration motor **10** can be mounted on, for example, a haptic device **200** schematically illustrated in FIG. **11**. The haptic device **200** includes a chassis **200A** and a vibration motor **10** accommodated in the chassis **200A**. The haptic device **200** is a device that gives a tactile stimulus to a person who operates the haptic device **200** by vibration of the vibration motor **10**.

The haptic device **200** illustrated in FIG. **11** is, for example, a stylus pen. Since the vibration motor **10** outputs vibration according to setting, it is possible to give haptic feedback to the operator as if the operator is operating the haptic device **200** on paper, a blackboard, or the like even though the operator is operating the haptic device **200** in contact with a tablet device or the like.

Note that the haptic device is not limited to a stylus pen, and a smartphone, a tablet, a game device, a wearable terminal, and the like can also be employed.

The example embodiments of the present disclosure have been described above. The scope of the present disclosure is not limited to the above example embodiment. The present disclosure can be implemented by making various changes to the above example embodiment without departing from the gist of the disclosure. The matters described in the above example embodiment can be optionally combined together, as appropriate, as long as there is no inconsistency.

As described above, the vibration motor **10** according to another aspect of the present disclosure includes the stationary portion **1** and the movable portion **5** capable of vibrating in the center axis direction with respect to the stationary portion **1**. The stationary portion includes the coil **2** to apply a driving force to the movable portion by energization, and the housing **3** that accommodates the movable portion and the coil. The housing has the notch **33** recessed from an outer edge of the housing toward the other side in the center axis direction with the outside of the housing as one side in the center axis direction. The movable portion includes the magnet **6A** and the holder **71** fixed to one side in the center axis direction of the magnet and having a larger radial width than the magnet. Assuming that a direction orthogonal to the center axis direction is a first direction, the holder has the holder notch **711** that is recessed from an outer edge of the holder to one side in the first direction and overlaps the notch as viewed from the first direction. The bottom portion **711A** of the holder notch is disposed on the other side in the first direction with respect to the outer edge of the magnet (first configuration).

In the first configuration, the holder notch **711** penetrates from one side in the center axis direction to the other side in the center axis direction (second configuration).

In the first or second configuration, the holder **71** includes the holder flat portion **714** that extends perpendicularly to the central axis, and the holder wall portion **715** that extends from the outer end portion in the first direction of the holder flat portion to the other side in the center axis direction. The holder flat portion covers at least a part of the outer end portion in the first direction of the magnet **6A**. The holder notch **711** is provided in the holder flat portion and the holder wall portion (third configuration).

The techniques, example embodiments, and modifications thereof, of the present disclosure, can be used for a vibration motor mounted on various devices, for example.

Features of the above-described example embodiments and the modifications thereof may be combined appropriately as long as no conflict arises.

While example embodiments of the present disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present disclosure. The scope of the present disclosure, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A vibration motor comprising:

a stationary portion; and

a movable portion to vibrate in a center axis direction with respect to the stationary portion; wherein

the stationary portion includes:

a coil to apply a driving force to the movable portion by energization; and

a housing that accommodates the movable portion and the coil;

the housing includes a notch recessed inward from an outer edge of the housing in the center axis direction; the notch includes:

a first notch recessed inward from an outer edge of the housing in the center axis direction; and

a second notch defined in an outer edge of the first notch and recessed inward from the outer edge of the first notch in the center axis direction; and

a lead wire of the coil extends out to an outside of the housing through the notch.

2. The vibration motor according to claim 1, wherein the housing includes:

a continuous portion continuously extending in a circumferential direction; and

a protruding portion protruding from one circumferential end portion of the continuous portion toward one side in the circumferential direction;

the second notch is provided in the protruding portion; and

the first notch is defined by the other circumferential end portion, the one circumferential end portion, and the protruding portion.

3. A haptic device comprising:

a chassis; and

the vibration motor according to claim 1 accommodated in the chassis.

4. The vibration motor according to claim 1, wherein the first notch and the second notch overlap one another when the vibration motor is viewed along the center axis direction.

5. The vibration motor according to claim 1, wherein the stationary portion includes a Flexible Printed Circuit board (FPC) fixed to the housing;

the housing includes a first region in which the first notch and the second notch are provided; and

the FPC is fixed to an outer peripheral surface of the first region so as to cover at least a portion of the first notch.

6. The vibration motor according to claim 5, wherein an outer side of the housing is defined as one side in the center axis direction;

the FPC includes an FPC notch recessed from an outer edge on another side in the center axis direction toward the one side in the center axis direction in a direction at a time of fixing to the first region; and

the lead wire of the coil extends out of the housing through the notch and the FPC notch.

7. The vibration motor according to claim 5, wherein the first region is a flat portion;

the housing includes a bent portion connected to both end portions of the first region in a circumferential direction; and

the FPC is fixed to the first region and the bent portion.

8. The vibration motor according to claim 7, wherein an outer side of the housing is defined as one side in the center axis direction;

the FPC includes:

a base portion extending in the center axis direction and fixed to the first region; and

a protruding piece that protrudes from the base portion to both sides in a circumferential direction and is fixed to the bent portion;

the base portion includes an electrode portion; and

the protruding piece overlaps the electrode portion in the center axis direction or is provided on the one side in the center axis direction with respect to the electrode portion as viewed in a direction perpendicular or substantially perpendicular to the first region.

\* \* \* \* \*